

# 1. Analytical Derivation of Sphere-of-Influence Encounter Points

## 1.1. Problem Statement

Given two Keplerian orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the anomaly parameters such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius  $r_{\text{SOI}}$ . The spacecraft orbit may be elliptic ( $e_1 < 1$ ) or hyperbolic ( $e_1 > 1$ ); the planet orbit is always elliptic.

$$\|A\mathbf{r}_1(E_1) - B\mathbf{r}_2(E_2)\| = r_{\text{SOI}} \quad (1)$$

where  $A$  and  $B$  are known constant  $3 \times 2$  rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

## 1.2. Notation

- $a_i, b_i, e_i$ : semi-major axis, semi-minor axis, and eccentricity of orbit  $i$  (with  $b_i = |a_i| \sqrt{1 - e_i^2}$ )
  - Elliptic ( $e_i < 1$ ):  $a_i > 0$
  - Hyperbolic ( $e_i > 1$ , spacecraft only):  $a_i < 0$
- $E_i$ : eccentric anomaly (elliptic orbit)
- $H_1$ : hyperbolic eccentric anomaly (hyperbolic spacecraft orbit)
- $A \in \mathbb{R}^{3 \times 2}, B \in \mathbb{R}^{3 \times 2}$ : orbital-to-ecliptic rotation matrices with components  $A_{jk}, B_{jk}$
- $r_{\text{SOI}}$ : radius of the planet's sphere of influence

## 1.3. Position in Orbital Plane

For an elliptic orbit, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\mathbf{r}_i(E_i) = \begin{bmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{bmatrix} \quad (2)$$

For a hyperbolic spacecraft orbit:

$$\mathbf{r}_1(H_1) = \begin{bmatrix} a_1(\cosh H_1 - e_1) \\ b_1 \sinh H_1 \end{bmatrix} \quad (3)$$

## 1.4. Distance Constraint

Square both sides of the constraint:

$$(A\mathbf{r}_1 - B\mathbf{r}_2)^\top (A\mathbf{r}_1 - B\mathbf{r}_2) = r_{\text{SOI}}^2 \quad (4)$$

Expand the left-hand side:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - 2\mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (5)$$

### 1.4.1. Orthonormality of $A$ and $B$

Since  $A$  and  $B$  are  $3 \times 2$  matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$A^\top A = I_{2 \times 2}, \quad B^\top B = I_{2 \times 2} \quad (6)$$

Therefore the self-terms simplify:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 = \mathbf{r}_1^\top \mathbf{r}_1 = |\mathbf{r}_1|^2 =: r_1^2 \quad (7)$$

$$\mathbf{r}_2^\top B^\top B \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_2 = |\mathbf{r}_2|^2 =: r_2^2 \quad (8)$$

### 1.4.2. The coupling matrix $C$

Define the  $2 \times 2$  matrix:

$$C := 2A^\top B \quad (9)$$

This encodes the entire mutual orientation of the two orbital planes. Equation 5 becomes:

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (10)$$

Note:  $C$  is **not** necessarily symmetric – it is a general  $2 \times 2$  real matrix with  $|C_{jk}| \leq 2$ .

## 1.5. Newton's Method

We seek to minimise  $f(E_1, E_2) := r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 - r_{\text{SOI}}^2$  using Newton's method. Each iteration solves the  $2 \times 2$  linear system  $H\delta = -\nabla f$  for the step  $\delta = (\delta E_1, \delta E_2)^\top$ , then updates  $E_i \leftarrow E_i + \delta E_i$ .

### 1.5.1. Deriving the gradient and Hessian

The objective is  $f = r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 - r_{\text{SOI}}^2$ , with  $r_{\text{SOI}}^2$  constant. We differentiate each part with respect to  $E_i$ .

#### 1.5.1.1. Self-terms $r_i^2$

Using  $r_i = a_i(1 - e_i \cos E_i)$ :

$$\frac{\partial r_i^2}{\partial E_i} = 2a_i^2 e_i \sin E_i (1 - e_i \cos E_i) \quad (11)$$

$$\frac{\partial^2 r_i^2}{\partial E_i^2} = 2a_i^2 e_i (\cos E_i - e_i + 2e_i \sin^2 E_i) \quad (12)$$

and  $\partial r_i^2 / \partial E_j = 0$  for  $i \neq j$ .

For a hyperbolic spacecraft orbit,  $r_1 = a_1(1 - e_1 \cosh H_1)$  (positive since  $a_1 < 0$  and  $1 - e_1 \cosh H_1 < 0$ ):

$$\frac{\partial r_1^2}{\partial H_1} = -2a_1^2 e_1 \sinh H_1 (1 - e_1 \cosh H_1) \quad (13)$$

$$\frac{\partial^2 r_1^2}{\partial H_1^2} = 2a_1^2 e_1 (e_1 - \cosh H_1 + 2e_1 \sinh^2 H_1) \quad (14)$$

#### 1.5.1.2. Cross-term $\mathbf{r}_1^\top C \mathbf{r}_2$

The position vector Equation 2 can be factored as  $\mathbf{r}_i = \text{diag}(a_i, b_i) \mathbf{p}_i$ , where  $\text{diag}(a_i, b_i) := \begin{pmatrix} a_i & 0 \\ 0 & b_i \end{pmatrix}$  is a diagonal matrix and  $\mathbf{p}_i$  is a dimensionless position-direction vector:

$$\mathbf{p}_i = \begin{bmatrix} \cos E_i - e_i \\ \sin E_i \end{bmatrix} \quad (15)$$

For a hyperbolic spacecraft orbit:  $\mathbf{p}_1 = (\cosh H_1 - e_1, \sinh H_1)^\top$ .

Substituting  $\mathbf{r}_i = \text{diag}(a_i, b_i) \mathbf{p}_i$  into the cross-term and using  $(X\mathbf{p})^\top = \mathbf{p}^\top X^\top$ :

$$\mathbf{r}_1^\top C \mathbf{r}_2 = (\text{diag}(a_1, b_1) \mathbf{p}_1)^\top C \text{diag}(a_2, b_2) \mathbf{p}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1)^\top C \text{diag}(a_2, b_2) \mathbf{p}_2 \quad (16)$$

Since diagonal matrices are symmetric ( $\text{diag}(x, y)^\top = \text{diag}(x, y)$ ):

$$\mathbf{r}_1^\top C \mathbf{r}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1) C \text{diag}(a_2, b_2) \mathbf{p}_2 \quad (17)$$

We have a chain of five matrices. Applying associativity ( $A(BC) = (AB)C$ ) twice – first to  $(\mathbf{p}_1^\top \cdot \text{diag}(a_1, b_1)) \cdot C = \mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot C)$ , then to  $(\mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot C)) \cdot \text{diag}(a_2, b_2) = \mathbf{p}_1^\top \cdot ((\text{diag}(a_1, b_1) \cdot C) \cdot \text{diag}(a_2, b_2))$  – lets us group the three constant middle factors into a single matrix. We define the **scaled coupling matrix**:

$$\mathbf{M} := \text{diag}(a_1, b_1) C \text{diag}(a_2, b_2) = \begin{pmatrix} C_{11}a_1a_2 & C_{12}a_1b_2 \\ C_{21}a_2b_1 & C_{22}b_1b_2 \end{pmatrix} \quad (18)$$

giving  $\mathbf{r}_1^\top C \mathbf{r}_2 = \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$ .

Since  $\mathbf{M}$  is constant, differentiation acts only on  $\mathbf{p}_1$  and  $\mathbf{p}_2$ . Define the derivative vectors:

$$\hat{\mathbf{w}}_i := \frac{d\mathbf{p}_i}{dE_i} = \begin{bmatrix} -\sin E_i \\ \cos E_i \end{bmatrix}, \quad \hat{\mathbf{u}}_i := \begin{bmatrix} \cos E_i \\ \sin E_i \end{bmatrix} \quad (19)$$

For a hyperbolic spacecraft orbit:

$$\hat{\mathbf{w}}_1 := \frac{d\mathbf{p}_1}{dH_1} = \begin{bmatrix} \sinh H_1 \\ \cosh H_1 \end{bmatrix}, \quad \hat{\mathbf{u}}_1 := \begin{bmatrix} \cosh H_1 \\ \sinh H_1 \end{bmatrix} \quad (20)$$

To differentiate  $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$  with respect to  $E_1$ , note that only  $\mathbf{p}_1$  depends on  $E_1$ . By associativity,  $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = \mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2)$ , which is a dot product of  $\mathbf{p}_1$  with the constant vector  $\mathbf{M} \mathbf{p}_2$ . The derivative of a dot product with one constant factor is:

$$\frac{\partial(\mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2))}{\partial E_1} = \dot{\mathbf{w}}_1^\top (\mathbf{M} \mathbf{p}_2) = \dot{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (21)$$

Similarly, for  $E_2$  only  $\mathbf{p}_2$  depends on  $E_2$ . By associativity,  $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = (\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2$ , a dot product of the constant row vector  $\mathbf{p}_1^\top \mathbf{M}$  with  $\mathbf{p}_2$ :

$$\frac{\partial((\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2)}{\partial E_2} = (\mathbf{p}_1^\top \mathbf{M}) \dot{\mathbf{w}}_2 = \mathbf{p}_1^\top \mathbf{M} \dot{\mathbf{w}}_2 \quad (22)$$

The same reasoning applies to higher derivatives, replacing  $\mathbf{p}_i$  by successive derivatives  $\dot{\mathbf{w}}_i$ ,  $-\dot{\mathbf{u}}_i$  (elliptic) or  $+\dot{\mathbf{u}}_i$  (hyperbolic, since  $d^2 \mathbf{p}_1 / dH_1^2 = +\dot{\mathbf{u}}_1$ ):

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_1^2} = -\dot{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2, \quad \frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_2^2} = -\mathbf{p}_1^\top \mathbf{M} \dot{\mathbf{u}}_2 \quad (23)$$

For a hyperbolic spacecraft orbit, the sign of the  $E_1$  term flips:

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial H_1^2} = +\dot{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (24)$$

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_1 \partial E_2} = \dot{\mathbf{w}}_1^\top \mathbf{M} \dot{\mathbf{w}}_2 \quad (25)$$

#### 1.5.1.3. Combined expressions

Combining  $f = r_1^2 + r_2^2 - \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 - r_{\text{SOI}}^2$  using Equation 11 with Equation 21, and Equation 12 with Equation 23 and Equation 25:

#### 1.5.2. Gradient

$$\frac{\partial f}{\partial E_1} = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \dot{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (26)$$

$$\frac{\partial f}{\partial E_2} = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M} \dot{\mathbf{w}}_2 \quad (27)$$

For a hyperbolic spacecraft orbit, Equation 26 becomes (using Equation 13 and the hyperbolic  $\dot{\mathbf{w}}_1$ ,  $\mathbf{p}_1$  from Equation 20):

$$\frac{\partial f}{\partial H_1} = -2a_1^2 e_1 \sinh H_1 (1 - e_1 \cosh H_1) - \dot{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (28)$$

Equation 27 is unchanged (the planet is always elliptic), but  $\mathbf{p}_1$  and  $\mathbf{M}$  use the hyperbolic definitions.

#### 1.5.3. Hessian

The Hessian  $\mathbf{H}$  is symmetric ( $H_{12} = H_{21}$ ):

$$H_{11} = 2a_1^2 e_1 (\cos E_1 - e_1 + 2e_1 \sin^2 E_1) + \dot{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (29)$$

$$H_{22} = 2a_2^2 e_2 (\cos E_2 - e_2 + 2e_2 \sin^2 E_2) + \mathbf{p}_1^\top \mathbf{M} \dot{\mathbf{u}}_2 \quad (30)$$

$$H_{12} = -\dot{\mathbf{w}}_1^\top \mathbf{M} \dot{\mathbf{w}}_2 \quad (31)$$

For a hyperbolic spacecraft orbit,  $H_{11}$  changes (using Equation 14 and Equation 24):

$$H_{11} = 2a_1^2 e_1 (e_1 - \cosh H_1 + 2e_1 \sinh^2 H_1) - \dot{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (32)$$

$H_{22}$  and  $H_{12}$  retain the same form, with the hyperbolic  $\dot{\mathbf{w}}_1$ ,  $\dot{\mathbf{u}}_1$ ,  $\mathbf{p}_1$ .

#### 1.5.4. Solving the Newton step

The  $2 \times 2$  system is solved directly via Cramer's rule:

$$\Delta = H_{11}H_{22} - H_{12}^2 \quad (33)$$

$$\delta E_1 = \frac{\frac{\partial f}{\partial E_1}H_{22} - \frac{\partial f}{\partial E_2}H_{12}}{-\Delta}, \quad \delta E_2 = \frac{\frac{\partial f}{\partial E_2}H_{11} - \frac{\partial f}{\partial E_1}H_{12}}{-\Delta} \quad (34)$$

All coupling terms are bilinear forms  $\mathbf{x}^\top \mathbf{M} \mathbf{y}$  with 2D vectors, each requiring 4 multiplies and 3 adds. The five needed dot products ( $\hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2$ ,  $\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2$ ,  $\hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2$ ,  $\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2$ ,  $\hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2$ ) can share the intermediate products  $\mathbf{M} \mathbf{p}_2$ ,  $\mathbf{M} \hat{\mathbf{w}}_2$ ,  $\mathbf{M} \hat{\mathbf{u}}_2$  (each a 2D matrix-vector multiply).

## 1.6. Coarse Encounter Interval

Before running Newton's method, we restrict the search to anomaly values where the spacecraft's heliocentric distance overlaps with the planet's distance range  $\pm r_{\text{SOI}}$ .

### 1.6.1. Radial overlap condition

An encounter requires the spacecraft distance to lie within:

$$a_2(1 - e_2) - r_{\text{SOI}} \leq r_1 \leq a_2(1 + e_2) + r_{\text{SOI}} \quad (35)$$

**Elliptic spacecraft.** Using  $r_1 = a_1(1 - e_1 \cos E_1)$  and solving for  $\cos E_1$ :

$$\frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \leq \cos E_1 \leq \frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \quad (36)$$

Clamping both sides to  $[-1, 1]$  and applying arccos (which reverses the inequality) gives two symmetric intervals  $[E_{\text{lo}}, E_{\text{hi}}]$  and  $[-E_{\text{lo}}, -E_{\text{hi}}]$  in  $E_1$ . If no valid interval exists (i.e. the clamped range is empty), no encounter is possible at any  $E_1$ .

**Hyperbolic spacecraft.** Using  $r_1 = a_1(1 - e_1 \cosh H_1)$  and solving for  $\cosh H_1$  (noting  $a_1 e_1 < 0$ , so dividing reverses the inequality):

$$\frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \leq \cosh H_1 \leq \frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \quad (37)$$

Clamping the lower bound to  $\max(\dots, 1)$  (since  $\cosh H_1 \geq 1$ ) and applying arcosh gives a single symmetric interval  $[-H_{\text{hi}}, -H_{\text{lo}}] \cup [H_{\text{lo}}, H_{\text{hi}}]$ . If the clamped range is empty, no encounter is possible.

### 1.6.2. Initial guess via parabolic interpolation

For each of the two symmetric  $E_1$  intervals, we evaluate  $f$  at three points: the endpoints and midpoint. Here  $E_2$  is estimated by projecting the spacecraft's 3D position onto the planet's orbital plane and inverting the ellipse parametrisation:

$$E_2 = \text{atan2}(q_y/b_2, q_x/a_2 + e_2) \quad (38)$$

where  $\mathbf{q} = \mathbf{B}^\top \mathbf{A} \mathbf{r}_1$  is the projection. This matches the eccentric anomaly that  $\mathbf{q}$  would have if it lay on the planet's ellipse. The projection is exact when the orbital planes coincide; for inclined orbits, the out-of-plane component is lost. Numerical evaluation for a Pluto-like target ( $i = 17^\circ$ ,  $e_2 = 0.25$ ) shows the maximum  $E_2$  error is under  $2^\circ$  for moderate spacecraft eccentricities ( $e_1 < 0.5$ ), but a direct Earth-Pluto near-Hohmann transfer ( $e_1 \approx 0.95$ ) reaches  $\sim 9^\circ$  at favourable orientations and up to  $\sim 63^\circ$  at unfavourable ascending-node orientations. Despite these large worst-case errors, the estimate remains adequate as an initial guess for Newton's method, which converges from any starting point in the correct half of the encounter region.

Given three samples  $(x_0, f_0)$ ,  $(x_1, f_1)$ ,  $(x_2, f_2)$ , the minimiser of the interpolating quadratic is:

$$E_1^* = x_1 - \frac{1}{2} \frac{(x_1 - x_0)^2(f_1 - f_2) - (x_1 - x_2)^2(f_1 - f_0)}{(x_1 - x_0)(f_1 - f_2) - (x_1 - x_2)(f_1 - f_0)} \quad (39)$$

Newton's method is then run from the best initial guess across both branches. If the first branch fails to converge to  $f \leq r_{\text{SOI}}^2$ , the second branch is tried.

### 1.7. Encounter Intervals in $E_1$ and $E_2$

The constraint  $f(E_1, E_2) = 0$  defines an implicit curve (or set of curves) in the  $(E_1, E_2)$  plane. The **encounter interval**  $[E_{1,\min}, E_{1,\max}]$  is the projection of this curve onto the  $E_1$  axis, and  $[E_{2,\min}, E_{2,\max}]$  is the projection onto the  $E_2$  axis. For every  $E_1$  in the first interval there exists at least one  $E_2$  in the second interval such that the distance equals  $r_{\text{SOI}}$ .

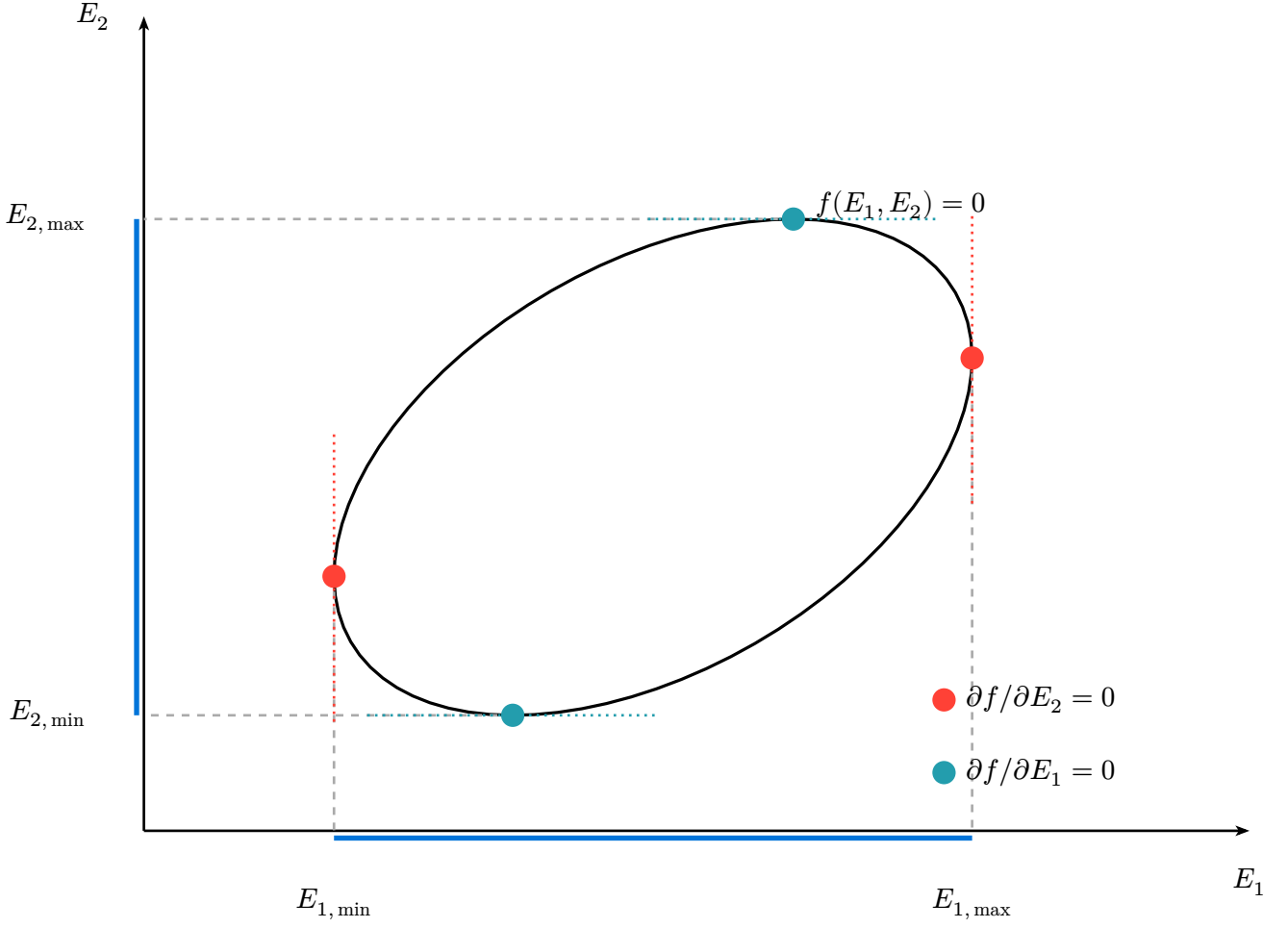


Figure 1: The constraint curve  $f = 0$  in the  $(E_1, E_2)$  plane. Red dots mark vertical tangents ( $\partial f / \partial E_2 = 0$ ), whose  $E_1$  coordinates give  $E_{1,\min}$  and  $E_{1,\max}$ . Teal dots mark horizontal tangents ( $\partial f / \partial E_1 = 0$ ), whose  $E_2$  coordinates give  $E_{2,\min}$  and  $E_{2,\max}$ . Blue bars on the axes show the projected encounter intervals.

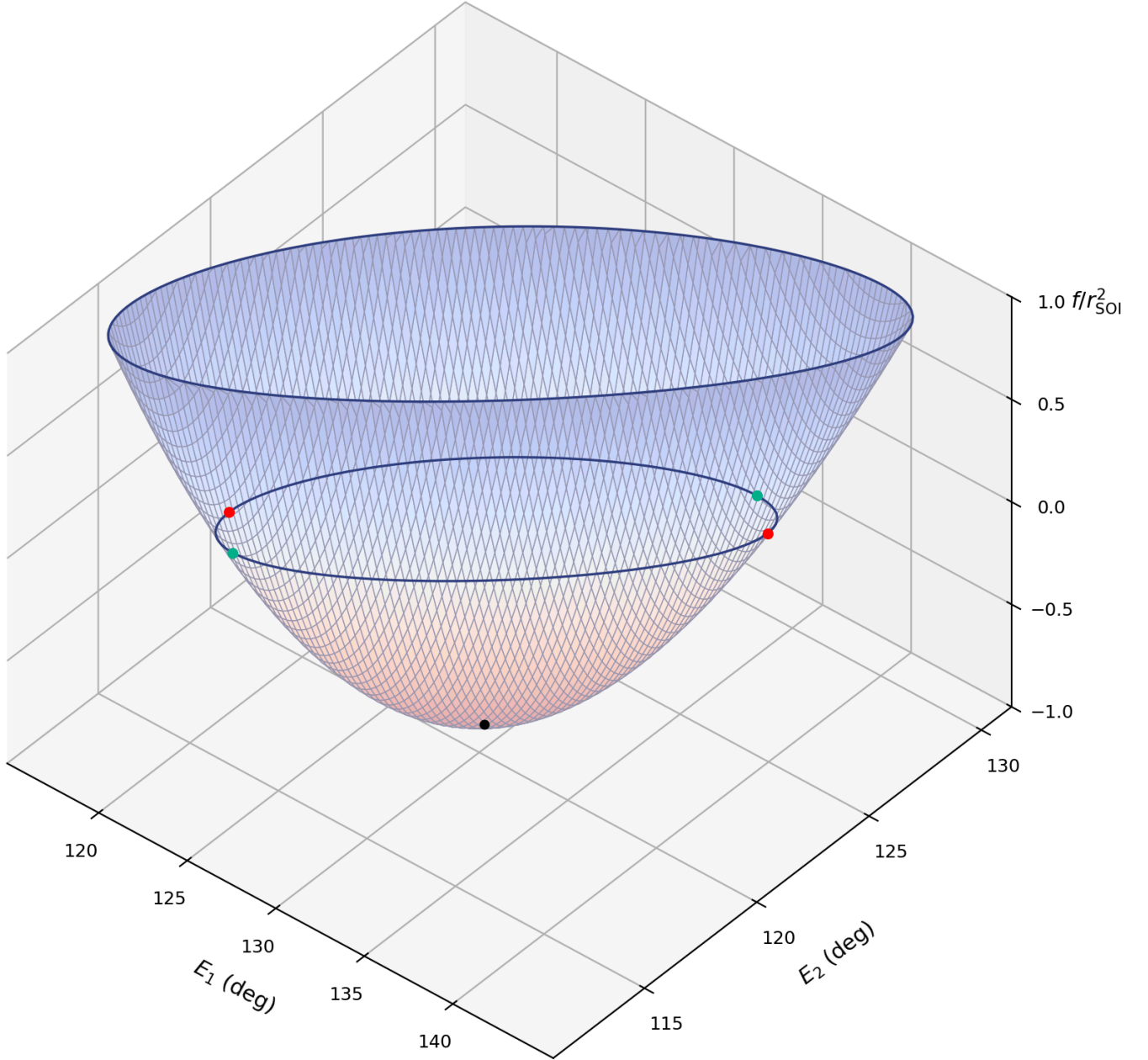


Figure 2: The surface  $z = f(E_1, E_2)/r_{\text{SOI}}^2$  over the encounter region (Voyager 2 – Jupiter). The black curve is the  $f = 0$  contour where the surface crosses the  $z = 0$  plane. The black dot at  $(E_1, E_2) \approx (128.9^\circ, 121.1^\circ)$  marks the surface minimum (closest approach without accounting for the planet’s gravity). Red dots mark  $\partial f/\partial E_2 = 0$  at  $E_1 \approx 120.2^\circ$  and  $139.0^\circ$  ( $E_1$  extrema); green dots mark  $\partial f/\partial E_1 = 0$  at  $E_2 \approx 115.3^\circ$  and  $126.9^\circ$  ( $E_2$  extrema).

### 1.7.1. Extrema via Lagrange multipliers

Finding the extreme values of  $E_2$  subject to  $f(E_1, E_2) = 0$  is a constrained optimisation problem. The Lagrangian is:

$$\mathcal{L} = E_2 - \lambda f(E_1, E_2) \quad (40)$$

Setting partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial E_1} = -\lambda \frac{\partial f}{\partial E_1} = 0, \quad \frac{\partial \mathcal{L}}{\partial E_2} = 1 - \lambda \frac{\partial f}{\partial E_2} = 0 \quad (41)$$

The second equation gives  $\lambda = 1/(\partial f/\partial E_2)$ , which is necessarily nonzero. Substituting into the first:

$$-\frac{1}{\partial f/\partial E_2} \cdot \frac{\partial f}{\partial E_1} = 0 \quad (42)$$

Since the prefactor is nonzero, this requires:

$$\frac{\partial f}{\partial E_1} = 0 \quad (43)$$

By the same argument with  $\mathcal{L} = E_1 - \lambda f$ , the extrema of  $E_1$  require:

$$\frac{\partial f}{\partial E_2} = 0 \quad (44)$$

Geometrically, consider the surface  $z = f(E_1, E_2)$  over the  $(E_1, E_2)$  plane (Figure 2). The constraint curve  $f = 0$  is the intersection of this surface with the  $z = 0$  plane. The  $E_2$  extrema of this intersection occur where the curve runs parallel to the  $E_1$  axis — at these points, the surface's gradient in the  $E_1$  direction is zero along the constraint, giving  $\partial f / \partial E_1 = 0$ .

### 1.7.2. System for $E_1$ bounds

To find  $E_{1, \min}$  and  $E_{1, \max}$ , solve the  $2 \times 2$  system:

$$\begin{cases} f(E_1, E_2) = 0 \\ \frac{\partial f}{\partial E_2} = 0 \end{cases} \quad (45)$$

Using Equation 27, the second equation is:

$$2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 = 0 \quad (46)$$

Newton's method for this system uses the Jacobian:

$$\mathbf{J}_1 = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial^2 f}{\partial E_1 \partial E_2} & \frac{\partial^2 f}{\partial E_2^2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ H_{12} & H_{22} \end{pmatrix} \quad (47)$$

where  $g_1 := \partial f / \partial E_1$  (Equation 26),  $g_2 := \partial f / \partial E_2$  (Equation 27), and  $H_{12}, H_{22}$  are the Hessian entries (Equation 31, Equation 30). The Newton step  $\boldsymbol{\delta} = (\delta E_1, \delta E_2)^\top$  solves  $\mathbf{J}_1 \boldsymbol{\delta} = -(\mathbf{f}, g_2)^\top$  via Cramer's rule:

$$\Delta_1 = g_1 H_{22} - g_2 H_{12} \quad (48)$$

$$\delta E_1 = \frac{f H_{22} - g_2^2}{\Delta_1}, \quad \delta E_2 = \frac{f H_{12} - g_1 g_2}{\Delta_1} \quad (49)$$

Note that at convergence ( $g_2 = 0$ ), the determinant simplifies to  $\Delta_1 = g_1 H_{22}$ .

### 1.7.3. System for $E_2$ bounds

To find  $E_{2, \min}$  and  $E_{2, \max}$ , solve:

$$\begin{cases} f(E_1, E_2) = 0 \\ \frac{\partial f}{\partial E_1} = 0 \end{cases} \quad (50)$$

Using Equation 26, the second equation is:

$$2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 = 0 \quad (51)$$

The Jacobian is:

$$\mathbf{J}_2 = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial^2 f}{\partial E_1^2} & \frac{\partial^2 f}{\partial E_1 \partial E_2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ H_{11} & H_{12} \end{pmatrix} \quad (52)$$

The Newton step solves  $\mathbf{J}_2 \boldsymbol{\delta} = -(\mathbf{f}, g_1)^\top$ :

$$\Delta_2 = g_1 H_{12} - g_2 H_{11} \quad (53)$$

$$\delta E_1 = \frac{f H_{12} - g_1 g_2}{\Delta_2}, \quad \delta E_2 = \frac{g_1^2 - f H_{11}}{\Delta_2} \quad (54)$$

At convergence ( $g_1 = 0$ ), the determinant simplifies to  $\Delta_2 = -g_2 H_{11}$ .

## 2. Time Constraint via Kepler's Equation

The preceding sections treated  $E_1$  and  $E_2$  as independent variables. In reality, both bodies obey Kepler's equation, which couples each eccentric anomaly to a common time  $t$ .

### 2.1. Kepler's equation

The mean motion  $n_i = \sqrt{\mu/|a_i|^3}$  applies to both elliptic and hyperbolic orbits, since the derivation from the vis-viva equation depends only on  $|a_i|$ .

For an elliptic orbit  $i$  with mean anomaly at epoch  $M_{i,0}$ :

$$M_i(t) = M_{i,0} + n_i t = E_i - e_i \sin E_i \quad (55)$$

For a hyperbolic spacecraft orbit:

$$M_1(t) = M_{1,0} + n_1 t = e_1 \sinh H_1 - H_1 \quad (56)$$

At a given time  $t$ , both anomalies are determined:

$$E_1 - e_1 \sin E_1 = M_{1,0} + n_1 t, \quad E_2 - e_2 \sin E_2 = M_{2,0} + n_2 t \quad (57)$$

(replacing the first equation with Equation 56 for a hyperbolic spacecraft).

### 2.2. Eliminating time

The position on an ellipse is  $2\pi$ -periodic in  $E_i$ , so each body returns to the same point after each full orbit. The set of times at which elliptic orbit  $i$  passes through eccentric anomaly  $E_i \in [-\pi, \pi]$  is:

$$t = \frac{E_i - e_i \sin E_i - M_{i,0} + 2\pi k_i}{n_i}, \quad k_i \in \mathbb{Z} \quad (58)$$

A hyperbolic orbit is not periodic — the spacecraft passes through each  $H_1$  exactly once:

$$t = \frac{e_1 \sinh H_1 - H_1 - M_{1,0}}{n_1} \quad (59)$$

For an encounter, both bodies must be at their respective positions **simultaneously**. For two elliptic orbits, equating the time expressions:

$$\frac{E_1 - e_1 \sin E_1 - M_{1,0} + 2\pi k_1}{n_1} = \frac{E_2 - e_2 \sin E_2 - M_{2,0} + 2\pi k_2}{n_2} \quad (60)$$

Rearranging gives a family of **time-coupling constraints**, one for each integer pair  $(k_1, k_2)$ :

$$h_{k_1, k_2}(E_1, E_2) := n_2(E_1 - e_1 \sin E_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) + 2\pi(n_2 k_1 - n_1 k_2) = 0 \quad (61)$$

Each choice of  $(k_1, k_2)$  corresponds to a different encounter opportunity (i.e. orbit 1 on its  $k_1$ -th revolution meeting orbit 2 on its  $k_2$ -th revolution). Since only the combination  $n_2 k_1 - n_1 k_2$  appears, the distinct constraints are parametrised by a single offset:

$$h(E_1, E_2; \alpha) := n_2(E_1 - e_1 \sin E_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) + 2\pi\alpha = 0 \quad (62)$$

where  $\alpha = n_2 k_1 - n_1 k_2$  ranges over a discrete set. In the  $(E_1, E_2)$  plane (with  $E_i \in [-\pi, \pi]$ ), each value of  $\alpha$  gives a monotone curve from bottom-left to top-right, and consecutive curves are spaced apart by one spacecraft orbital period  $T_1 = 2\pi/n_1$ . The full encounter problem is: find  $(E_1, E_2)$  satisfying both the distance constraint  $f = 0$  (Equation 10) and  $h = 0$  (Equation 62) for some admissible  $\alpha$ .

**Hyperbolic spacecraft.** Since  $k_1 = 0$  (Equation 59), the time constraint simplifies to a family parametrised by  $k_2$  alone. Replacing the spacecraft's mean anomaly expression:

$$h(H_1, E_2; k_2) := n_2(e_1 \sinh H_1 - H_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) - 2\pi n_1 k_2 = 0 \quad (63)$$

### 2.3. Finding the next encounter after $t_0$

After each gravity assist, the spacecraft's orbit changes, so it only makes sense to find the **next** encounter from a given epoch  $t_0$  rather than enumerating all future encounters on a fixed orbit.

The distance constraint can produce up to two separate encounter zones (corresponding to the two symmetric coarse intervals from Equation 36). Each zone  $s \in \{1, 2\}$  has its own encounter intervals  $[E_{1,\min}^{(s)}, E_{1,\max}^{(s)}]$  and  $[E_{2,\min}^{(s)}, E_{2,\max}^{(s)}]$  from Figure 1. Since the mean anomaly  $M_i = E_i - e_i \sin E_i$  is monotone in  $E_i$ , each zone corresponds to mean-anomaly intervals:

$$M_{i,\min}^{(s)} = E_{i,\min}^{(s)} - e_i \sin E_{i,\min}^{(s)}, \quad M_{i,\max}^{(s)} = E_{i,\max}^{(s)} - e_i \sin E_{i,\max}^{(s)} \quad (64)$$

For a hyperbolic spacecraft orbit:  $M_{1,\min}^{(s)} = e_1 \sinh H_{1,\min}^{(s)} - H_{1,\min}^{(s)}$  (and likewise for  $M_{1,\max}^{(s)}$ ).

On its  $k_i$ -th elliptic orbit, body  $i$  passes through encounter zone  $s$  during the time interval (Equation 58):

$$t \in \left[ \frac{M_{i,\min}^{(s)} - M_{i,0} + 2\pi k_i}{n_i}, \frac{M_{i,\max}^{(s)} - M_{i,0} + 2\pi k_i}{n_i} \right] \quad (65)$$

A hyperbolic spacecraft passes through each encounter zone at most once. Setting  $k_1 = 0$  in the above (with the hyperbolic mean anomaly from Equation 59):

$$t \in \left[ \frac{M_{1,\min}^{(s)} - M_{1,0}}{n_1}, \frac{M_{1,\max}^{(s)} - M_{1,0}}{n_1} \right] \quad (66)$$

An encounter requires both bodies to be in their respective intervals of the **same** zone **simultaneously**.

#### 2.3.1. Choosing the iteration order

Label the body with the tighter encounter zone (smaller  $M_{i,\max}^{(s)} - M_{i,\min}^{(s)}$ ) as  $i$  and the other as  $j$ . Iterating over passes of the tighter-zone body produces fewer candidates, since each pass occupies a shorter time interval and is less likely to overlap with the other body. The formulas below use  $i$  for the outer loop and  $j$  for the overlap check. For a hyperbolic spacecraft, always start with the spacecraft ( $i = 1$ ): it has a single fixed time interval per zone (Equation 66), so just find which planet passes ( $k_2$ ) overlap it.

#### 2.3.2. First pass of body $i$ after $t_0$

For each zone  $s$ , body  $i$  enters the encounter zone for the first time after  $t_0$  on orbit number:

$$k_i^{(s)} = \left\lceil \frac{n_i t_0 + M_{i,0} - M_{i,\max}^{(s)}}{2\pi} \right\rceil \quad (67)$$

This is the smallest integer  $k_i$  for which the encounter interval of body  $i$  (Equation 65) has not ended before  $t_0$ . Start with  $k_i^* = \min_s k_i^{(s)}$ , the earliest pass that enters **either** zone.

#### 2.3.3. Checking for overlap with body $j$

On each pass  $k_i^*$ , check both zones  $s \in \{1, 2\}$ . The encounter window of body  $i$  in zone  $s$ , clamped to start no earlier than  $t_0$ , is:

$$t_{i,\text{lo}}^{(s)} = \max \left( \frac{M_{i,\min}^{(s)} - M_{i,0} + 2\pi k_i^*}{n_i}, t_0 \right), \quad t_{i,\text{hi}}^{(s)} = \frac{M_{i,\max}^{(s)} - M_{i,0} + 2\pi k_i^*}{n_i} \quad (68)$$

If  $t_{i,\text{lo}}^{(s)} > t_{i,\text{hi}}^{(s)}$ , body  $i$  is not in zone  $s$  on this pass. Otherwise, the earliest orbit of body  $j$  that could overlap is:

$$k_j = \left\lceil \frac{n_j t_{i,\text{lo}}^{(s)} + M_{j,0} - M_{j,\max}^{(s)}}{2\pi} \right\rceil \quad (69)$$

Body  $j$  enters zone  $s$  on orbit  $k_j$  at time  $t_{j,\text{lo}} = \frac{M_{j,\min}^{(s)} - M_{j,0} + 2\pi k_j}{n_j}$ . If  $t_{j,\text{lo}} \leq t_{i,\text{hi}}^{(s)}$ , the two windows overlap — this zone produces a candidate encounter. If both zones produce a match on the same pass, pick the one with the smaller  $t_{j,\text{lo}}$  (the

earlier encounter). The resulting pair  $(k_1, k_2)$  yields  $\alpha = n_2 k_1 - n_1 k_2$ , which is the value to use in Newton's method on the combined system (Equation 72).

If neither zone produces a match on pass  $k_i^*$ , increment  $k_i^* \leftarrow k_i^* + 1$  and repeat. For a hyperbolic spacecraft ( $i = 1$ ), there is no incrementing — if no planet pass overlaps the spacecraft's single encounter window, no encounter exists on this orbit.

## 2.4. Gradient of $h$

$$\frac{\partial h}{\partial E_1} = n_2(1 - e_1 \cos E_1), \quad \frac{\partial h}{\partial E_2} = -n_1(1 - e_2 \cos E_2) \quad (70)$$

For a hyperbolic spacecraft orbit:

$$\frac{\partial h}{\partial H_1} = n_2(e_1 \cosh H_1 - 1) \quad (71)$$

This is always positive (since  $e_1 > 1$  and  $\cosh H_1 \geq 1$ ), matching the sign of the elliptic case.

## 2.5. Newton's method on the combined system

For a given  $\alpha$ , we solve the  $2 \times 2$  system  $(f, h) = \mathbf{0}$  using Newton's method. The Jacobian is:

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial h}{\partial E_1} & \frac{\partial h}{\partial E_2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ n_2(1 - e_1 \cos E_1) & -n_1(1 - e_2 \cos E_2) \end{pmatrix} \quad (72)$$

The Newton step  $\delta = (\delta E_1, \delta E_2)^\top$  solves  $\mathbf{J}\delta = -(f, h)^\top$  via Cramer's rule:

$$\Delta = -g_1 n_1(1 - e_2 \cos E_2) - g_2 n_2(1 - e_1 \cos E_1) \quad (73)$$

$$\delta E_1 = \frac{f n_1(1 - e_2 \cos E_2) + g_2 h}{\Delta} \quad (74)$$

$$\delta E_2 = \frac{-f n_2(1 - e_1 \cos E_1) - g_1 h}{\Delta} \quad (75)$$

where  $g_1, g_2$  are the distance-constraint gradient components (Equation 26, Equation 27) and  $h$  is evaluated from Equation 62 (so  $\alpha$  enters the Newton step through  $h$ ). Since  $\partial h / \partial E_1 > 0$  and  $\partial h / \partial E_2 < 0$  always, the Jacobian is non-singular whenever  $(g_1, g_2)$  is not parallel to  $(n_2 r_1 / a_1, -n_1 r_2 / a_2)$ .

For a hyperbolic spacecraft orbit, the same Cramer's rule formulas apply with  $(1 - e_1 \cos E_1)$  replaced by  $(e_1 \cosh H_1 - 1)$  (Equation 71),  $g_1$  from Equation 28, and  $h$  from Equation 63.



### 2.6.2. Linearised time constraint

The time constraint  $h(E_1, E_2; \alpha) = 0$  is linearised around  $(E_1^*, E_2^*)$ :

$$h^* - h_1^* \delta_1 - h_2^* \delta_2 = 0 \quad (79)$$

where  $h^* = h(E_1^*, E_2^*; \alpha)$ ,  $h_1^* = n_2(1 - e_1 \cos E_1^*)$ ,  $h_2^* = -n_1(1 - e_2 \cos E_2^*)$  (Equation 70). This gives  $\delta_2 = c + m\delta_1$  with:

$$c = \frac{h^*}{h_2^*}, \quad m = \frac{h_1^*}{h_2^*} \quad (80)$$

### 2.6.3. Intersection

Substituting  $\delta_2 = c + m\delta_1$  into Equation 78 yields a quadratic in  $\delta_1$ :

$$A\delta_1^2 + B\delta_1 + C = 0 \quad (81)$$

with coefficients:

$$A = H_{11} + 2H_{12}m + H_{22}m^2 \quad (82)$$

$$B = 2c(H_{12} + H_{22}m) \quad (83)$$

$$C = H_{22}c^2 + 2f_{\min} \quad (84)$$

The two roots correspond to the two intersections of the time-constraint line with the encounter ellipse. The initial guess is the root with the larger  $\delta_1$  (smaller  $E_1$ , earlier encounter):

$$\delta_1 = \frac{-B + \sqrt{B^2 - 4AC}}{2A} \quad (85)$$

giving  $(E_1^{(0)}, E_2^{(0)}) = (E_1^* - \delta_1, E_2^* - c - m\delta_1)$ . If the discriminant  $B^2 - 4AC < 0$ , the time-constraint line does not intersect the encounter ellipse for this  $\alpha$ , and this encounter opportunity can be skipped.